CSE10(8263)

Simulation & Modeling of a Shared Dormitory Laundry Room

Annika Lois R. Dumalogdog

Agenda

Problem Definition & Objectives

Model Design & Structure

Simulation Setup & Execution

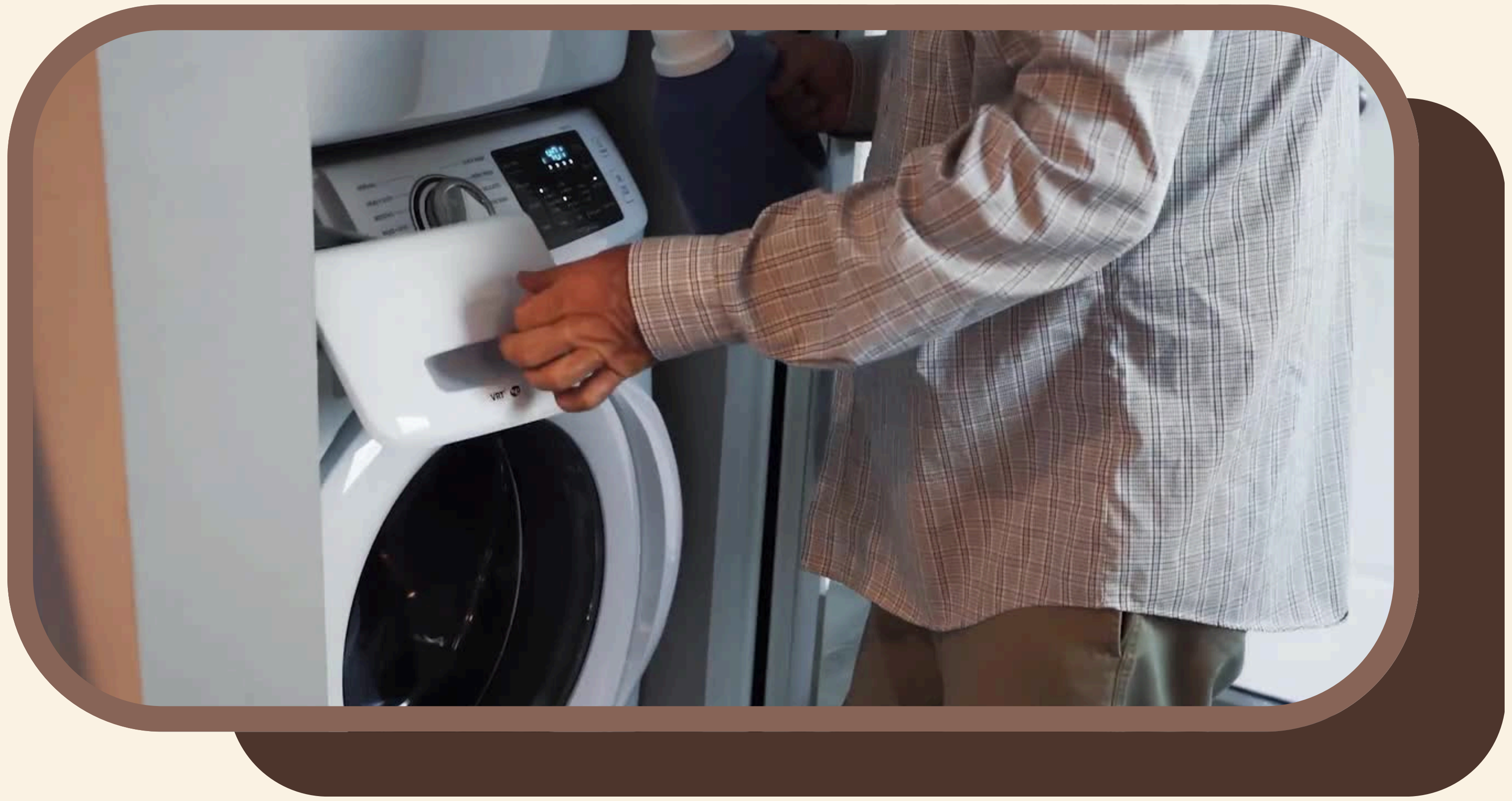
Data Usage & Analysis

Recommendations & Conclusion

Presentation & Visual Flow

1.1 Problem Definition

Residents in shared housing complexes (dormitories, apartments, condominiums) frequently complain about long waits at the communal laundry room. The problem is amplified by two compounding factors: Laundry demand is not uniform, and Machine hogging, where users often leave clothes inside machines long after the wash cycle ends, blocking the next person in the queue.



1.2 Objectives

- To model a dormitory laundry room queuing system using discrete-event simulation.
- To measure average resident wait time, machine utilization rate, and peak queue length across four machine-count scenarios
- To visualize the laundry room system dynamics using a web-based program animation showing real-time queue formation and machine status.
- To provide actionable recommendations for dormitory management on machine investment and operational policy changes.

2.1 System Description

The dormitory laundry room system consists of the following key components that interact to serve residents:

Entities

Residents from a non-homogeneous Poisson process

Resources

N washing machines (1–4 configurable)

Queues

FIFO per machine, capacity 8 each

Event

Arrival, Start Wash, End Wash, Hog, Release, Balk

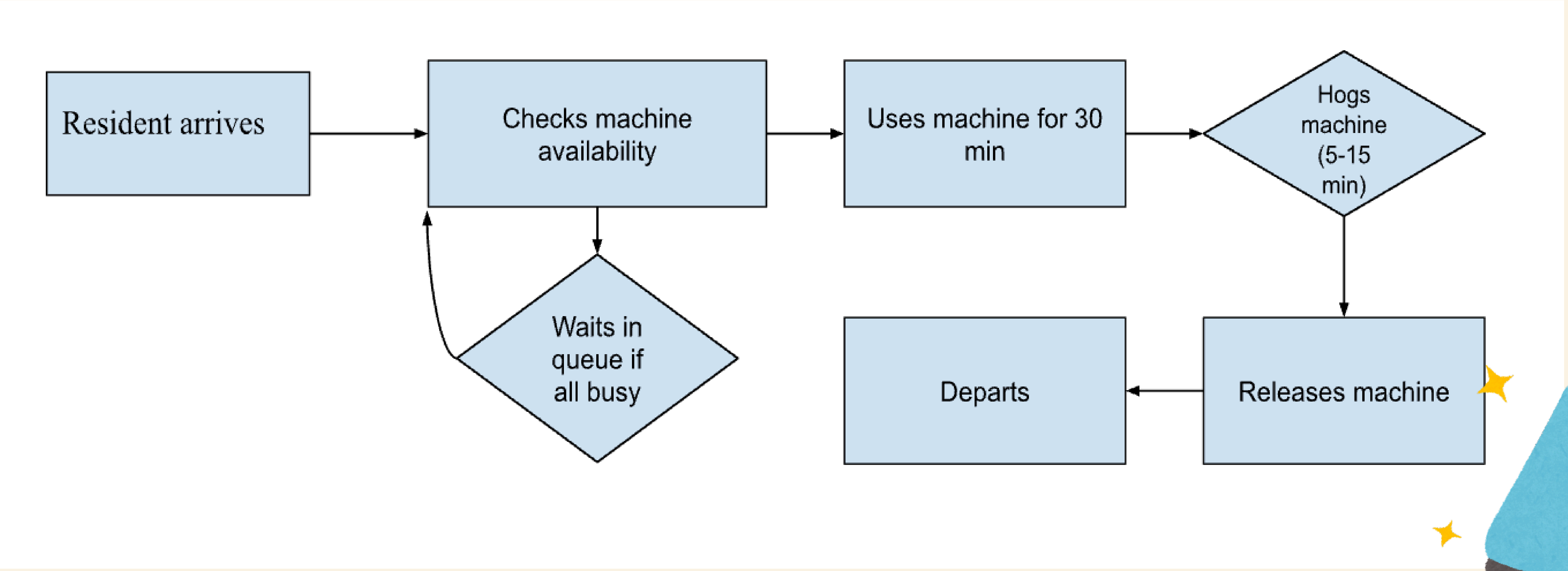
Attributes

arrivalTime, startWashTime, endWashTime, hogEndTime

Run Length

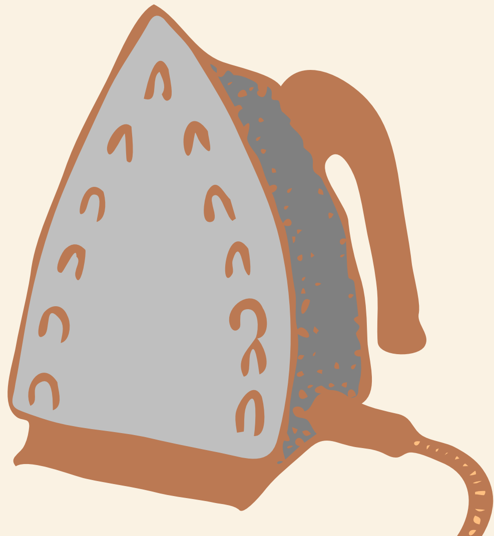
960 min (16-hour day) per replication

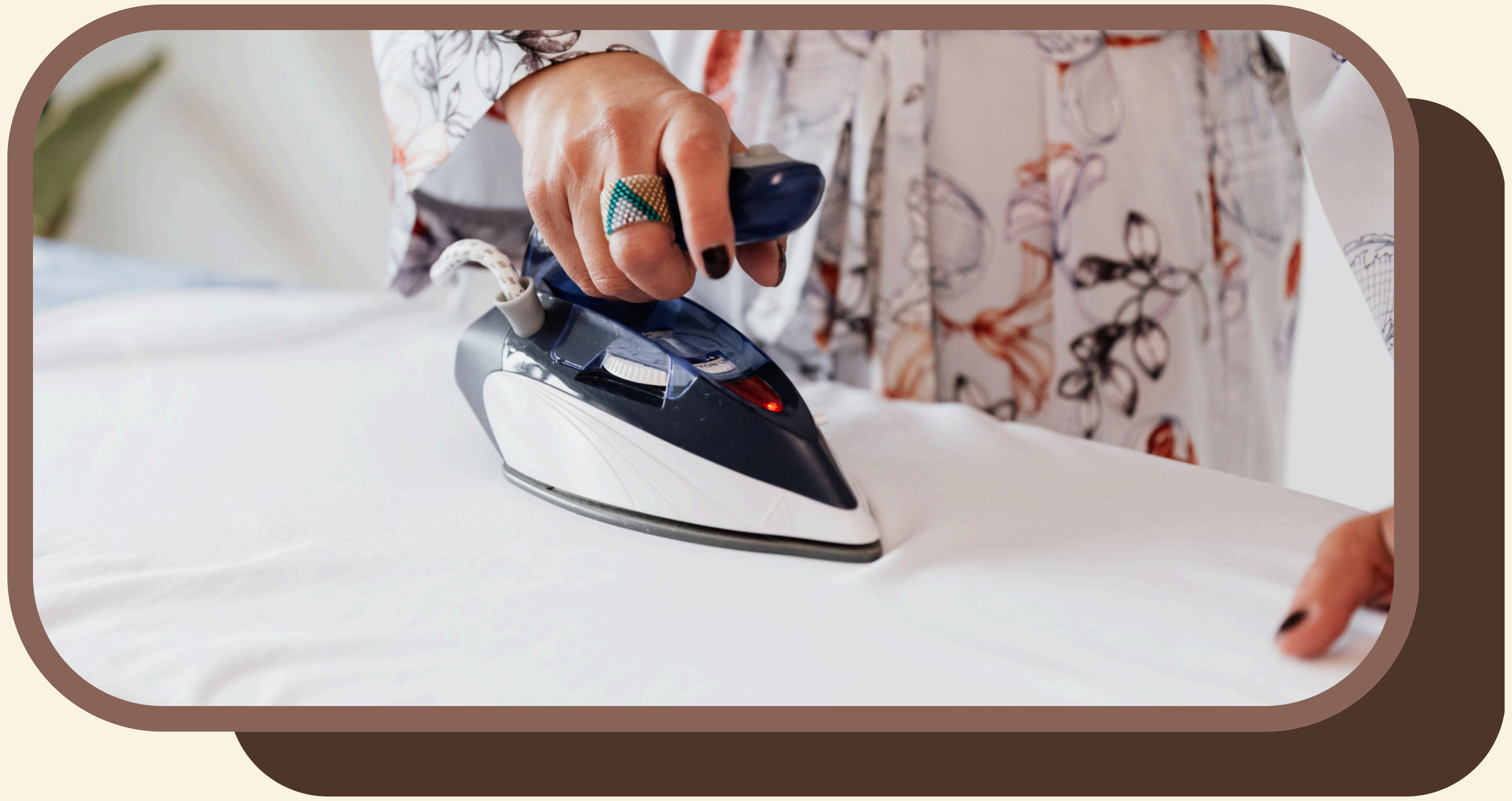
2.2 Module Structure



2.3 Modeling Assumptions

1. **Arrivals:** Non-homogeneous Poisson with hourly λ(t)
2. **Wash Time:** Uniform, (25-35) minutes
3. **Hog Probability:** p = 0.3 after every wash
4. **Hog Duration:** Uniform, (5-15) minutes
5. **Queue Policy:** FIFO per machine, capacity = 8
6. **Routing:** Join the shortest queue, no jockeying
7. **Balking:** Full system → resident leaves and never returns
8. **Machines:** Identical, no failures or maintenance





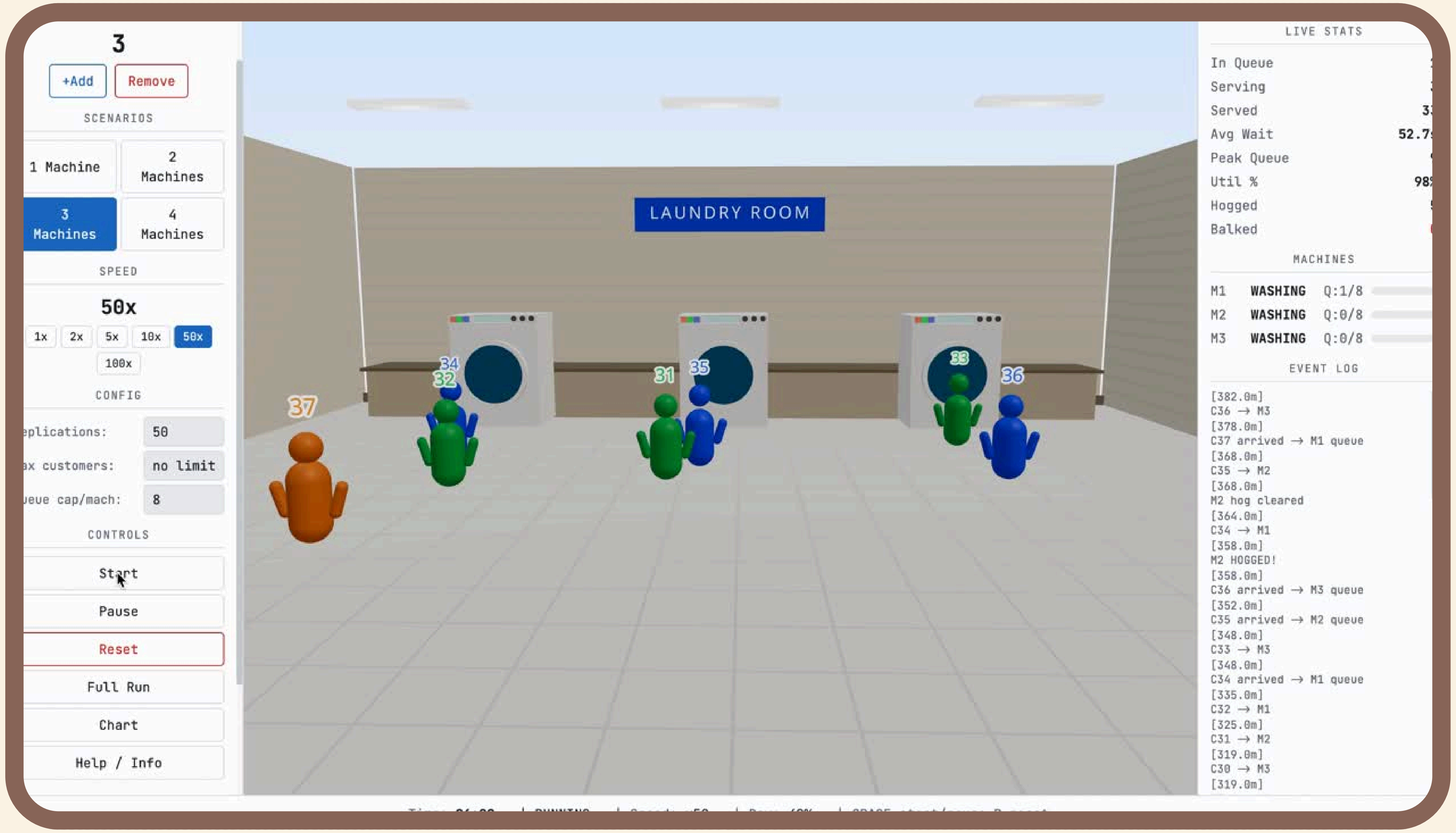
3.1 Parameters & execution

KPIs Measured: Average wait time, Peak queue length, Machine utilization %, Served customers, Balked customers, and Hogged events

Parameters:

- Run Length: 960 min (16-hour day)
- Replication:50 independent runs
- Random Seeds: Independent per replication
- Configurations: 1, 2, 3, 4 machines
- Queue Capacity: 8 per machine

3.2 Execution and GUI



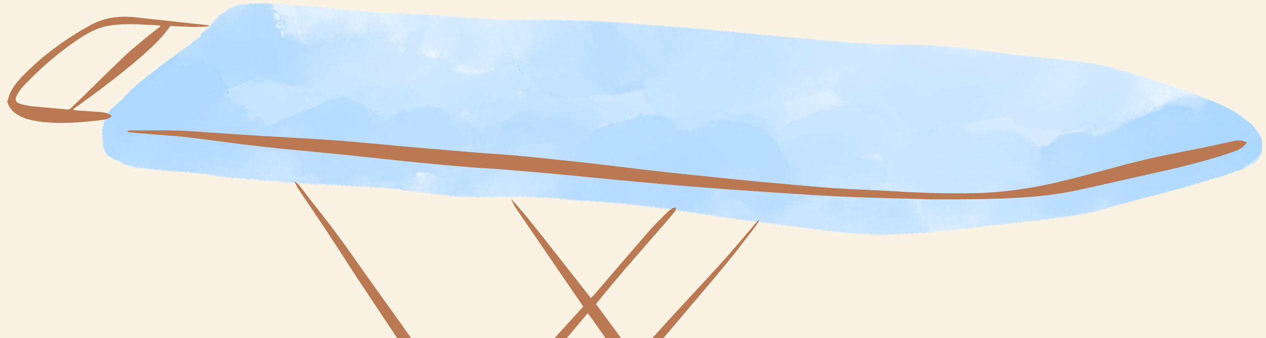
3D Scene
React Three Fiber renders machines & residents in real time.

Speed control
1×, 2×, 5×, 10×, 50×, 100× to inspect or fast-forward.

Live KPIs
Wait, utilization, peak queue update every tick.

Batch Mode
50 replications run instantly for statistical analysis.

Charts
Performance charts and CSV export after each run.



4.1 Results

Mean ± half-width of 95% CI across 50 replications, 8 queue cap

Machines	Avg Wait (min)	Utilization %	Peak Queue	Served	Hogged	Balked %
1	202.16	99.29	8	28.38	8.82	49.12
2	183.58	98.85	15.92	56.42	17.38	17.12
3	52.60	91.5	14.36	78.48	22.92	0
4	26.17	72.87	10.2	83.36	24.84	0

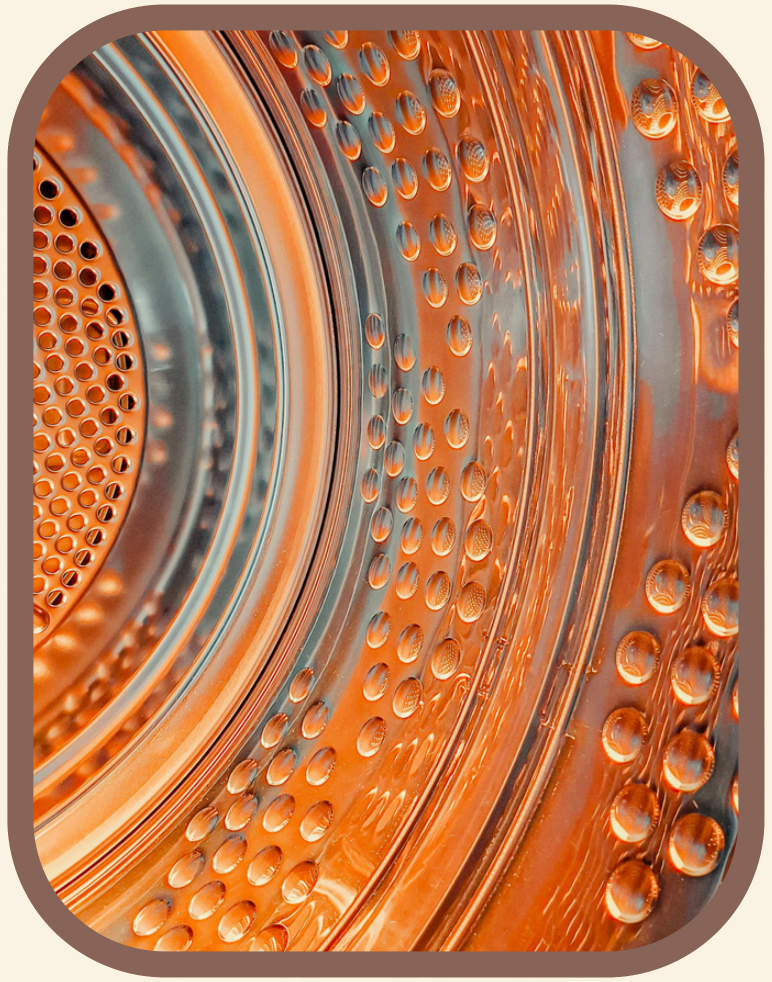
Recommended

3 Machines

It's the smallest configuration where balking hits zero, and the system stays stable.
Best cost-to-service ratio.

4.2 Analysis

1 machine is severely overloaded, utilization 99%, queues permanently saturated, ~49% balking.



2 machines cut wait time by 20min but peak queues still reach 15 residents

3 machines are the most optimal: 52-min wait, 91% utilization, zero balking

4 machines deliver near-instant service, but utilization drops to ~30%, which is over-provisioned

5.1 Recommendations

DEPLOY 3 MACHINES
Optimal trade-off: less than 1-hour wait, no balking, healthy utilization.

ANTI-HOG POLICY
Notifications when cycle ends; small fee after 10 min idle.

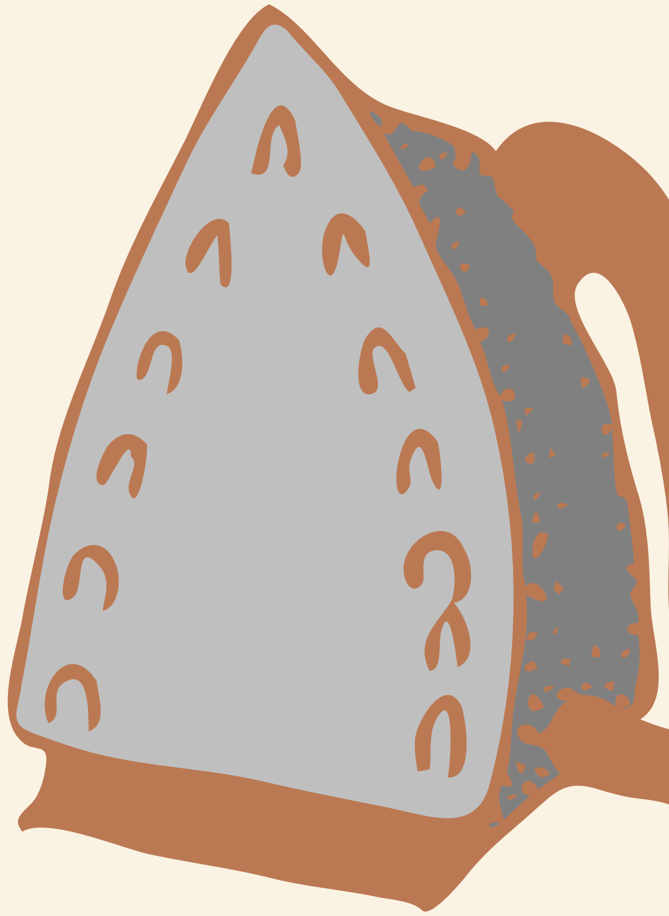
PUBLIC DASHBOARD
Live machine-status display reduces perceived wait and complaints.

RE-RUN ANNUALLY
Refresh inputs as occupancy or habits change to confirm 3 machines remain enough.

This project demonstrated how Discrete-Event Simulation can analyze a shared dormitory laundry room without disrupting real operations. The model captured time-varying arrivals, finite queue capacity, and realistic hogging behavior, and was validated through 50 replications per configuration with 95% confidence intervals.

Results show that increasing the number of machines dramatically reduces wait times and eliminates queueing entirely at the cost of lower utilization. Management should choose 3 machines since it shows promising results, while the management remains cost-efficient with the number of machines they will buy in their space.

5.2 Conclusion



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Simulation

6.1. Presentation and Usability